PROJECT ASSIGNMENT & SPECIFICATIONS

E-COMMERCE BUSINESS INTELLIGENCE: SALES, CUSTOMER & PROFITABILITY ANALYSIS

Comprehensive SQL Analytics Project Specification for Junior Analysts

Document Title:	E-commerce Business Intelligence: Sales, Customer & Profitability Analysis
Prepared For:	Junior Data Analyst
Project Sponsor:	Head of Business Analytics
Database Engine:	MySQL
Estimated Duration:	14 Days
Project Type:	Business Intelligence & SQL Analytics

2. Database Design

The project utilizes a normalized relational database named **"ecommerce_analytics"**. The schema comprises six interconnected tables designed to capture the complete lifecycle of e-commerce transactions, from customer signup to order fulfillment, payment settlement, and potential returns.

TABLE NAME	PRIMARY KEY	KEY FOREIGN KEYS	CORE PURPOSE
customers	customer_id	None	Stores demographic and registration details for all registered shoppers.
orders	order_id	customer_id	Records order-level header data, placement timestamps, and fulfillment status.
order_items	order_item_id	order_id, product_id	Line-item details linking orders to specific products, quantities, and transaction prices.
products	product_id	None	Catalog repository containing product names, categories, and baseline cost prices.
payments	payment_id	order_id	Tracks transaction amounts, payment methods, and gateway clearance status.
returns	return_id	order_id, product_id	Logs product returns, return timestamps, and customer-stated reasons.

Detailed Schema Specifications

- **customers:** `customer_id` (INT, PK), `customer_name` (VARCHAR), `city` (VARCHAR), `signup_date` (DATE).
- **orders:** `order_id` (INT, PK), `customer_id` (INT, FK → customers), `order_date` (DATETIME), `status` (ENUM/VARCHAR: Completed, Cancelled, Pending).
- **order_items:** `order_item_id` (INT, PK), `order_id` (INT, FK → orders), `product_id` (INT, FK → products), `quantity` (INT), `unit_price` (DECIMAL).
- **products:** `product_id` (INT, PK), `product_name` (VARCHAR), `category` (VARCHAR), `cost_price` (DECIMAL).
- **payments:** `payment_id` (INT, PK), `order_id` (INT, FK → orders), `payment_method` (VARCHAR), `amount` (DECIMAL), `payment_status` (VARCHAR: Success, Failed, Pending, Refunded).
- **returns:** `return_id` (INT, PK), `order_id` (INT, FK → orders), `product_id` (INT, FK → products), `return_date` (DATE), `reason` (VARCHAR).

3. Business Questions

Your analysis must directly address thirty specific business questions organized across four core analytical domains. For each question, you must construct the appropriate SQL query, verify data integrity, and interpret the output.

A. Sales and Revenue Analysis

1. Total Order Volume

How many orders were placed during the analysis period? Purpose: Establish baseline transaction volume.

2. Total Sales Value

What is the total sales value (quantity × unit_price for non-cancelled orders)? Purpose: Measure top-line revenue.

3. Average Order Value (AOV)

What is the average order value? Purpose: Assess customer basket size and purchasing power.

4. Peak and Trough Sales Months

Which months had the highest and lowest sales? Purpose: Identify seasonality and demand fluctuations.

5. Sales by Product Category

Which product categories generated the highest sales? Purpose: Direct merchandising and inventory investment.

6. Geographic Sales Distribution

Which cities generated the most sales? Purpose: Guide regional marketing and logistics placement.

7. Order Fulfillment Status Breakdown

How many orders were completed, cancelled, or pending? Purpose: Monitor operational bottlenecks and fulfillment health.

8. Top 10 Products by Sales Value

What are the top 10 products by sales value? Purpose: Identify core revenue drivers.

9. Monthly Sales Trend Analysis

How has monthly sales performance changed over time? Purpose: Track macro growth trajectories.

B. Customer Analytics

10. Top 10 Customers by Spending

Who are the top 10 customers by total spending? Purpose: Identify VIP clients for loyalty programs.

11. High-Frequency Shoppers

Which customers placed more than five orders? Purpose: Isolate core loyal customer cohorts.

12. Customer Geographic Concentration

Which cities have the largest customer base? Purpose: Inform geographic expansion strategies.

13. Monthly Customer Acquisition

How many new customers signed up each month? Purpose: Track marketing acquisition channel efficiency.

14. Repeat Purchase Rate

What percentage of customers made repeat purchases? Purpose: Measure customer stickiness and satisfaction.

15. Dormant Customer Identification

Which customers have not purchased in the last 90 days? Purpose: Design targeted win-back campaigns.

16. Highest Average Order Value Customers

Which customers have the highest average order value? Purpose: Identify high-ticket buyers.

C. Product and Profitability Analysis

17. Top Revenue-Generating Products

Which products generate the highest sales? Purpose: Reorder prioritization.

18. Estimated Gross Profit by Product

Which products generate the highest estimated gross profit (Sales minus Cost)? Purpose: Focus on bottom-line contribution.

19. Gross Profit Margins by Category

Which categories have the highest gross profit margins? Purpose: Strategic portfolio optimization.

20. High-Sales, Low-Margin Paradox

Which products have high sales but low profit margins? Purpose: Re-evaluate pricing or supplier costs.

21. Product Return Rates

Which products have the highest return rates (Returns / Quantity Sold)? Purpose: Identify defective items or listing discrepancies.

22. Primary Return Drivers

What are the most common product return reasons? Purpose: Address root causes of customer dissatisfaction.

D. Payment and Operational Analysis

23. Order Cancellation Rate

What percentage of orders were cancelled? Purpose: Evaluate fulfillment friction.

24. Payment Method Adoption

Which payment methods are most frequently used? Purpose: Optimize checkout gateway options.

25. Payment Status Valuation

What is the monetary value of successful, failed, pending, and refunded payments? Purpose: Financial reconciliation.

26. Declining Sales Categories

Which product categories show declining sales over recent quarters? Purpose: Flag category fatigue.

27. Top Products per Category

What are the top three products within each category (using window functions)? Purpose: Merchandising hierarchy insights.

28. Segment Contribution to Total Sales

Which customer segments contribute the most to total sales? Purpose: Pareto analysis (80/20 rule verification).

29. Month-over-Month Growth Rate

What is the month-over-month sales growth rate (using LAG)? Purpose: Dynamic growth tracking.

30. Critical Business Problems

What are the five most important business problems revealed by the analysis? Purpose: Executive synthesis.

4. SQL Skills to Cover

To successfully resolve the business questions, your SQL scripts must demonstrate proficiency across foundational and advanced querying concepts in MySQL:

- **Data Retrieval & Filtering:** `SELECT`, `WHERE`, `ORDER BY`, `LIMIT`, `DISTINCT`, and robust `NULL` handling (`COALESCE`, `IFNULL`).
- **Aggregation:** `COUNT`, `SUM`, `AVG`, `MAX`, `MIN` combined with `GROUP BY` and `HAVING` clauses.

- **Conditional Logic:** `CASE WHEN` statements for segmentation, categorization, and conditional metrics.
- **Table Joins:** `INNER JOIN`, `LEFT JOIN`, and multi-table relational stitching across all six schema entities.
- **Subqueries & CTEs:** Scalar/correlated subqueries and modular Common Table Expressions (`WITH` clause).
- **Window Functions:** Advanced analytical ranking and offset calculations using `ROW_NUMBER()`, `RANK()`, `DENSE_RANK()`, and `LAG()`.
- **String & Date Functions:** Date arithmetic (`DATE_ADD`, `DATEDIFF`, `DATE_FORMAT`) and string manipulation.
- **Database Management:** Creating reusable `VIEWS`, defining primary/foreign key `CONSTRAINTS`, and managing `TRANSACTIONS` (`START TRANSACTION`, `COMMIT`, `ROLLBACK`).

5. Data Cleaning Requirements

Real-world data is inherently messy. Before executing analytical queries, you must perform a comprehensive data auditing and cleaning protocol:

- **Duplicate Detection:** Identify and flag duplicate records across primary keys and transaction logs using `GROUP BY ... HAVING COUNT(*) > 1`.
- **Missing Value Audits:** Scan for `NULL` values in critical fields (e.g., customer city, product cost price, payment amounts) and apply imputation or exclusion rules.
- **Sanity Checks on Financials:** Detect invalid negative prices, zero or negative quantities, and mismatched payment totals against order line items.
- **Relational Integrity:** Validate foreign key constraints to ensure no orphaned order items exist without a corresponding valid order or product.
- **Standardization:** Clean inconsistent text formats (e.g., casing in customer cities, divergent order statuses such as 'Complete' vs 'Completed').
- **Date Validation:** Ensure chronological consistency (e.g., order date is after customer signup date; return date is after order date).

Note: Original raw tables must remain unaltered. All cleaning operations and data transformation steps must be documented in a separate script or markdown note.

6. Standardized Metric Definitions

To ensure consistency across executive dashboards and reports, adhere strictly to the following mathematical definitions:

METRIC NAME	CALCULATION FORMULA	BUSINESS INTERPRETATION
Total Sales Value	∑ (Quantity × Unit Price) for all non-cancelled order items	Gross revenue generated from product sales.
Average Order Value (AOV)	Total Sales Value ÷ Total Completed Orders	Average monetary spend per transaction.

METRIC NAME	CALCULATION FORMULA	BUSINESS INTERPRETATION
Estimated Gross Profit	∑ [Quantity × (Unit Price – Cost Price)]	Absolute profit contribution before operating expenses.
Gross Profit Margin	(Estimated Gross Profit ÷ Total Sales Value) × 100	Percentage of revenue retained as profit.
Product Return Rate	(Total Units Returned ÷ Total Units Sold) × 100	Proportion of purchased merchandise returned by customers.
Order Cancellation Rate	(Cancelled Orders ÷ Total Orders Placed) × 100	Measure of order fallout and checkout friction.
Repeat-Purchase Rate	(Customers with > 1 Order ÷ Total Unique Customers) × 100	Customer loyalty and retention indicator.
Month-over-Month Growth	[(Current Month Sales – Prior Month Sales) ÷ Prior Month Sales] × 100	Trajectory of revenue expansion.

7. Project Roadmap

Complete this assignment within a structured 14-day timeline. The schedule balances data preparation, technical querying, and strategic interpretation.

TIMELINE	PHASE / FOCUS AREA	KEY TASKS & SQL CONCEPTS	EXPECTED DELIVERABLE
Day 1	Project Kickoff	Review requirements, schema design, and business context.	Project plan & environment setup.
Day 2	Database Setup	Create MySQL database, define tables, keys, and constraints.	`schema.sql` script.
Day 3	Data Cleaning	Audit duplicates, missing values, date formats, and anomalies.	`data_cleaning.sql` script & notes.
Days 4–5	Sales Analysis	Write queries for Q1–Q9 (revenue, trends, geography).	`sales_analysis.sql` script.
Days 6–7	Customer Analytics	Write queries for Q10–Q16 (cohorts, retention, LTV).	`customer_analysis.sql` script.
Days 8–9	Profitability & Returns	Write queries for Q17–Q22 (gross profit, margins, returns).	`profitability_returns.sql`.
Days 10–11	Advanced SQL	Write queries for Q23–Q30 (window functions, CTEs, MoM growth).	`advanced_analysis.sql`.
Day 12	Insights & Strategy	Translate query outputs into business recommendations.	Draft executive report.

TIMELINE	PHASE / FOCUS AREA	KEY TASKS & SQL CONCEPTS	EXPECTED DELIVERABLE
Day 13	Documentation	Organize GitHub repository, write README and comments.	Structured repository.
Day 14	Portfolio Review	Final polish, code review, and portfolio preparation.	Final submission package.

8. Final Deliverables & Repository Structure

Upon completion, organize your project work into a professional GitHub repository following this standardized structure:

```
ecommerce_analytics/
│
├── README.md                          # Comprehensive project documentation
├── sql/
│   ├── 01_schema.sql                  # Database creation and table definitions
│   ├── 02_data_cleaning.sql           # Data auditing and anomaly correction
│   ├── 03_sales_analysis.sql          # Queries for Questions 1-9
│   ├── 04_customer_analysis.sql       # Queries for Questions 10-16
│   ├── 05_profitability_returns.sql   # Queries for Questions 17-22
│   └── 06_advanced_analysis.sql       # Queries for Questions 23-30 (CTEs, Window Functions)
├── reports/
│   ├── business_recommendations.md    # Executive summary and strategic insights
│   └── metrics_definitions.md         # Documentation of calculation methodologies
└── dashboard/
    └── powerbi_preview.png            # Optional dashboard wireframe or export
```

9. Translating Insights to Strategic Recommendations

A data analyst's value is measured not just by clean code, but by actionable business impact. When converting your SQL query outputs into recommendations for management, structure every finding using the following four-part framework:

- **Finding:** What factual pattern or anomaly did the SQL query reveal?
- **Supporting Metric:** What exact data point or percentage substantiates the finding?
- **Business Impact:** Why does this matter to the company's bottom line or operational efficiency?
- **Suggested Action:** What specific operational, marketing, or merchandising step should management take?

> **Hypothetical Example**
>
> **Finding:** High-revenue product category suffers from disproportionate return rates.
> **Supporting Metric:** Category X accounts for 35% of total sales but exhibits a 14.2% return rate (company average is 4.5%).
> **Business Impact:** Margin erosion due to reverse logistics costs and customer dissatisfaction.

> **Suggested Action:** Audit supplier quality for Category X and update product page descriptions to set accurate customer expectations.